

Alert Analysis Report

Ransomware Attack Chain Investigation

PCAP Analysis – Week 10

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Date of Analysis:	April 6, 2026
Case Reference:	PCAP Analysis – Week 10
Infected Host:	172.16.1.67 (JERSEY-SHORE-1A) – Patient Zero
Organization:	CARLFORCE.COM – 172.16.1.0/24
Toolset:	Security Onion, Wireshark, VirusTotal, NetworkMiner
Malware Family:	Ransomware / RIG Exploit Kit / Dridex C2

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1. Abstract

This analysis investigates network traffic captured on December 23, 2017 from the fictional organization **CARLFORCE.COM**, operating on the internal subnet 172.16.1.0/24. The investigation confirms a multi-stage ransomware attack chain originating from internal host **172.16.1.67 (JERSEY-SHORE-1A)**, identified as Patient Zero.

The infection began with a drive-by download of a malicious Word document (`Invoice[1].doc`) from `fundtel.es`, followed by exploit kit payload delivery (`2042[1].exe`) from `enjoyveganrecipes.com` via the **RIG Exploit Kit**. Command-and-control infrastructure leveraged dynamic DNS providers including DynDNS and No-IP (`gabby4u.linkpc.net`), triggering **ET_DYN_DNS** Suricata alerts. Ransomware indicators including Tor-based payment pages (`topyzscsu5poprxy.onion.link`) and Bitcoin redirect services (`coinurl.com`) were observed across multiple internal hosts. A total of **four internal hosts** showed indicators of compromise within the 38-minute capture window, with lateral movement confirmed via dynamic DNS activity on a previously unaffected machine.

2. Introduction

The purpose of this investigation is to analyse the following Suricata IDS alerts generated in Security Onion:

- ET EXPLOIT_KIT RIG EK URI Struct Mar 13 2017
- ET EXPLOIT_KIT RIG EK URI Struct Mar 13 2017 M2
- ET EXPLOIT_KIT RIG EK URI Struct Oct 24 2016 (RIG-v)
- ET EXPLOIT_KIT RIG EK Landing Sep 12 2016 T2
- ET_DYN_DNS Observed DNS Query to DynDNS Domain (`linkpc.net`)
- ET_INFO DynDNS CheckIp External IP Address Server Response
- ET_INFO External IP Lookup -- `checkip.dyndns.org`

These alerts indicate a multi-stage attack beginning with a malicious Word document dropper, progressing through the RIG Exploit Kit, and culminating in ransomware deployment with active Tor-based payment infrastructure. The objective is to reconstruct the full attack sequence, identify all indicators of compromise, and assess the likelihood of accurate malware identification versus false positives.

3. Methodology

- Reviewed the Security Onion Alerts dashboard to identify and correlate Suricata rule triggers by source IP, destination IP, and rule name.
- Expanded individual alerts to examine full packet payloads, rule metadata, confidence ratings, and MITRE ATT&CK mappings.
- Loaded `Week10.pcap` into Wireshark and applied display filters to isolate traffic per malicious IP and reconstruct HTTP streams.
- Used Wireshark display filters (`http.host` and `http.request.method == "POST"`) to identify drive-by download traffic, exploit kit payload delivery, and C2 beaconing sessions.

- Opened **Week10.pcap** in NetworkMiner to enumerate hosts, extract files, identify OS fingerprints, and review outgoing sessions from the infected host.
- Queried each IOC (IP addresses and domains) in VirusTotal to obtain independent vendor verdicts and community intelligence.
- Cross-referenced all timestamps, IP addresses, and hostnames to verify consistency and build a chronological attack timeline.

4. Findings

4.1 Environment

NetworkMiner identified the following internal network topology from the PCAP:

Table 1. Network Environment – CARLFORCE.COM

Field	Value
PCAP File	Week10.pcap
Capture Date	2017-12-23
Capture Window	01:05:49 – 01:43:38 UTC (37 min 49 sec)
Total Packets	8,159
Internal Network	172.16.1.0/24
Domain Controller	172.16.1.4 (CARLFORCE-DC1)
Patient Zero	172.16.1.67 (JERSEY-SHORE-1A, Windows 7)
Additional Hosts	172.16.1.89, 172.16.1.141, 172.16.1.201
Toolset	Wireshark, NetworkMiner 2.8.1, Security Onion 2.4.160, Virus-Total

4.2 Patient Zero Identification

NetworkMiner's Hosts tab confirmed **172.16.1.67 (JERSEY-SHORE-1A)** as Patient Zero. Key indicators include a dual user-agent fingerprint — a legitimate IE11 string and a spoofed IE7 string associated with Dridex malware — a JA3 hash matching the Tofsee malware family, and the first malicious HTTP request in the capture originating from this host with 58 outgoing sessions recorded.

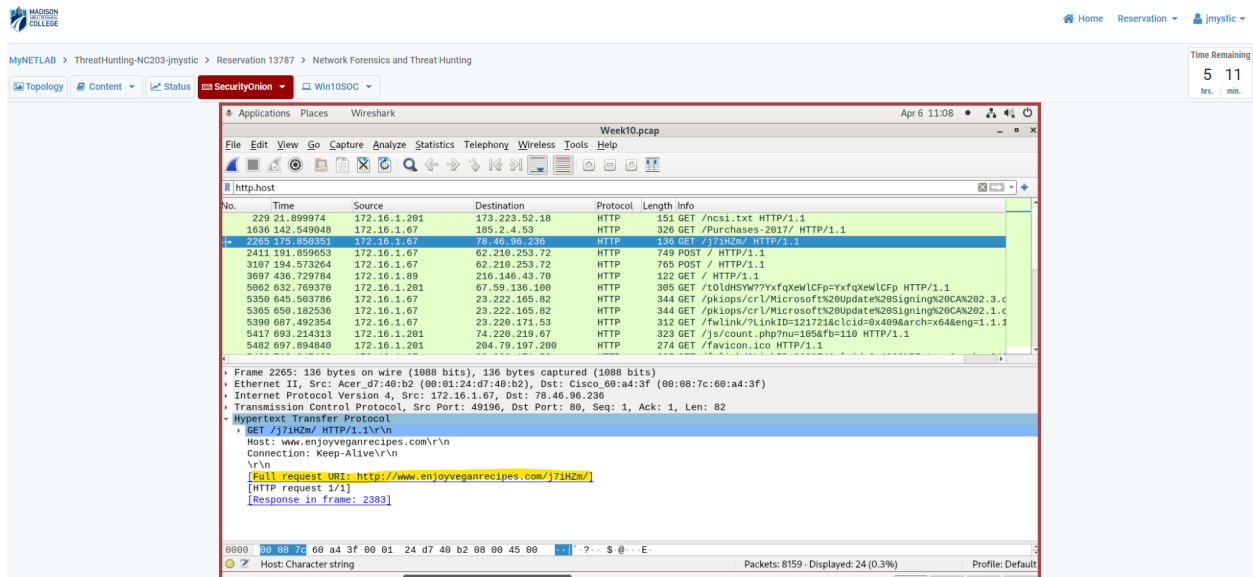


Figure 1. Figure 1: NetworkMiner Hosts tab – 172.16.1.67 (JERSEY-SHORE-1A) identified as Patient Zero. Dual user-agents highlighted in yellow (legitimate IE11 and spoofed IE7 Dridex UA). JA3 hash matches Tofsee malware.

4.3 Malicious Files Extracted

NetworkMiner's Files tab extracted both malicious files directly from the PCAP, confirming the initial access and exploit kit delivery stages:

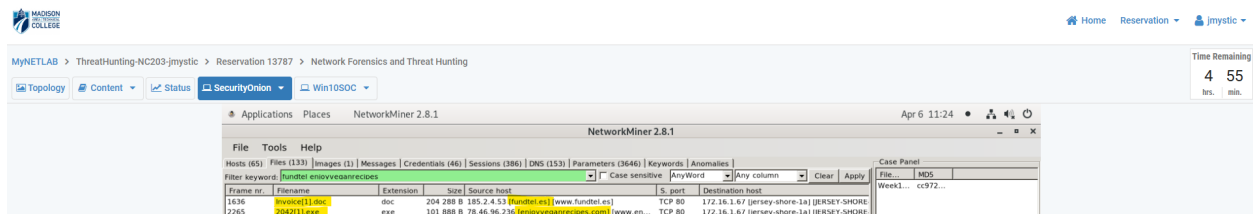


Figure 2. Figure 2: NetworkMiner Files tab – Invoice[1].doc (204 KB) from fundtel.es and 2042[1].exe (101 KB) from enjoyveganrecipes.com, both delivered to 172.16.1.67 (JERSEY-SHORE-1A).

4.4 Stage 1 – Initial Access: Drive-by Download

At approximately **01:08 UTC**, host **172.16.1.67** issued an HTTP GET request to **www.fundtel.es/Purchases-2017/**. The server at **185.2.4.53** responded with an **application/msword** file — the malicious Word document **Invoice[1].doc** (204 KB). The filename “Invoice” is a social engineering lure designed to trick the user into opening the document and enabling macros.

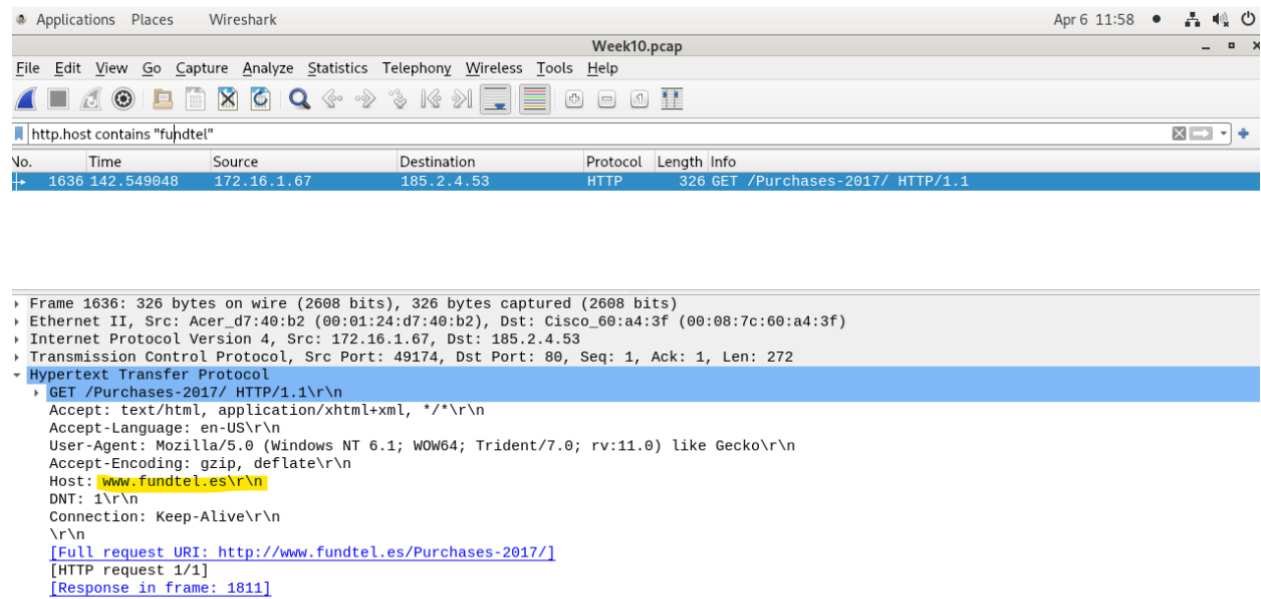


Figure 3. Figure 3: Wireshark – HTTP GET /Purchases-2017/ from 172.16.1.67 to 185.2.4.53 (fundtel.es). Full request URI highlighted. Frame 1636, Time 142.549 s.

4.5 Stage 2 – Exploit Kit Payload Delivery (RIG EK)

Following the Word document download, **172.16.1.67** issued a second HTTP GET to **www.enjoyveganrecipes.com/j7iHZm/**. The server at **78.46.96.236** returned an **application/octet-stream** — the malware executable **2042[1].exe** (101 KB). This two-stage pattern is characteristic of the **RIG Exploit Kit**, confirmed by Security Onion's Suricata alert **ET EXPLOIT_KIT RIG EK URI Struct Mar 13 2017** (SID 2024048, severity: High).

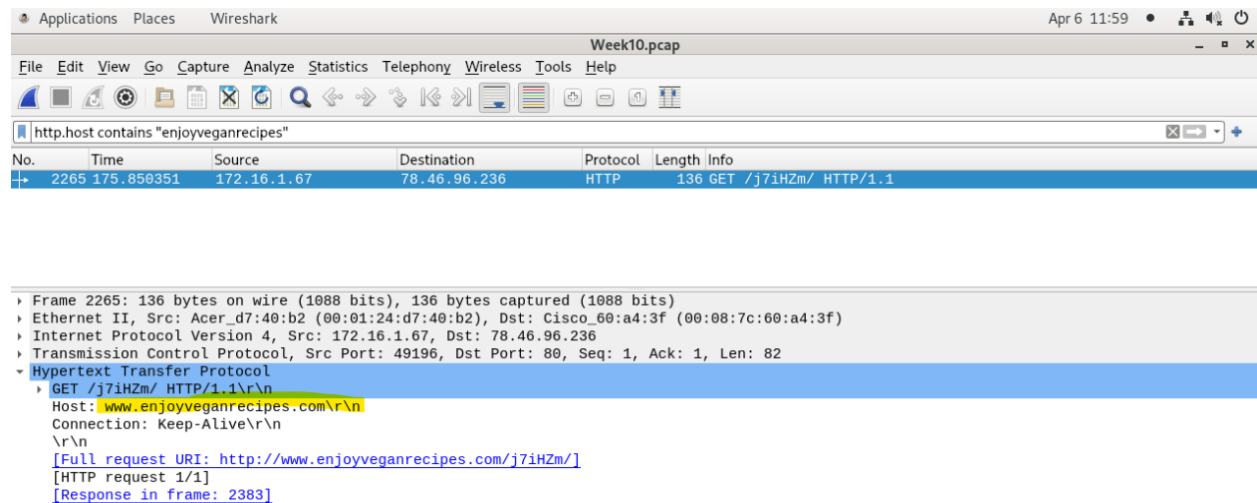


Figure 4. Figure 4: Wireshark – HTTP GET /j7iHZm/ from 172.16.1.67 to 78.46.96.236 (enjoyveganrecipes.com). Full request URI highlighted. Frame 2265, Time 175.850 s.

4.6 Stage 3 – C2 Beaconing

Host **172.16.1.67** sent repeated HTTP POST requests to **62.210.253.72:8080** beginning at 01:12 UTC using a hardcoded spoofed user-agent string consistent with **Dridex** malware. The actual browser on the machine was IE11, making the IE7 declaration a definitive detection indicator. Additional C2 sessions were observed to **220.227.247.35:7080**, **220.227.247.39:7080**, and **153.149.153.41:8080**, indicating a resilient multi-server C2 architecture.

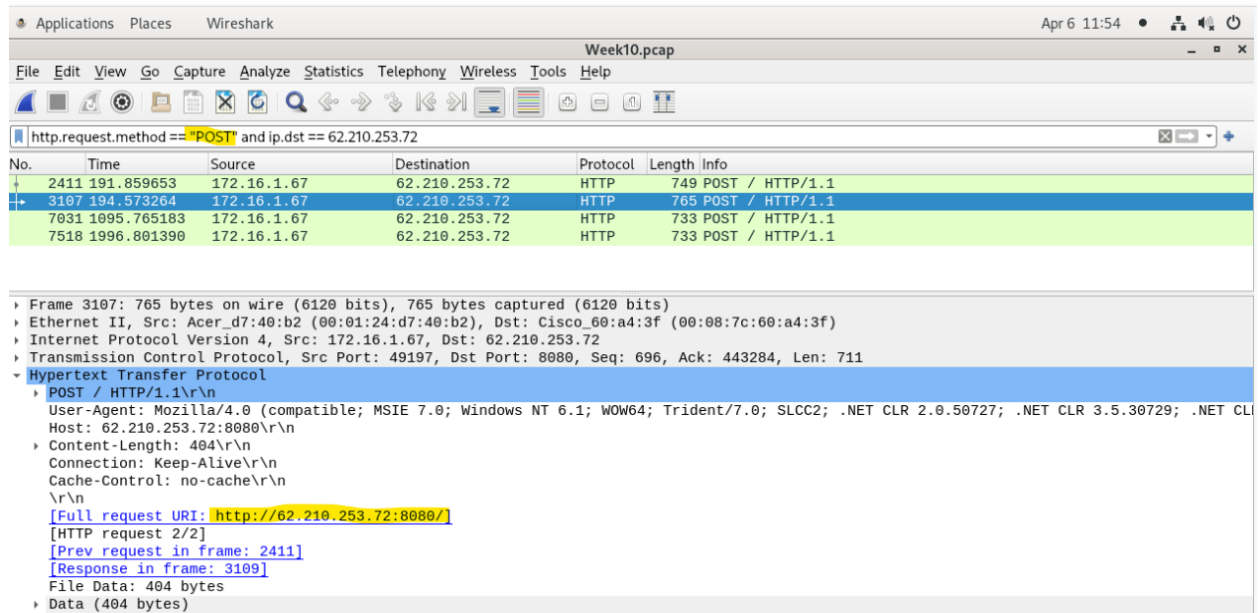


Figure 5. Figure 5: Wireshark – Repeated HTTP POST requests from 172.16.1.67 to 62.210.253.72:8080. Filter: `http.request.method == "POST"` and `ip.dst == 62.210.253.72`. Four POST packets displayed with spoofed IE7 user-agent.

4.7 Stage 4 – ET_DYN_DNS Alerts

checkip.dyndns.org (172.16.1.89): At 01:13 UTC, host **172.16.1.89** queried `checkip.dyndns.org`. The DynDNS server at 216.146.43.70 responded with the host's external IP address. This is a known malware technique to confirm internet connectivity before C2 contact. Suricata fired ET INFO External IP Lookup -- `checkip.dyndns.org` (SID 2021378) and ET INFO DynDNS CheckIp External IP Address Server Response (SID 2014932).

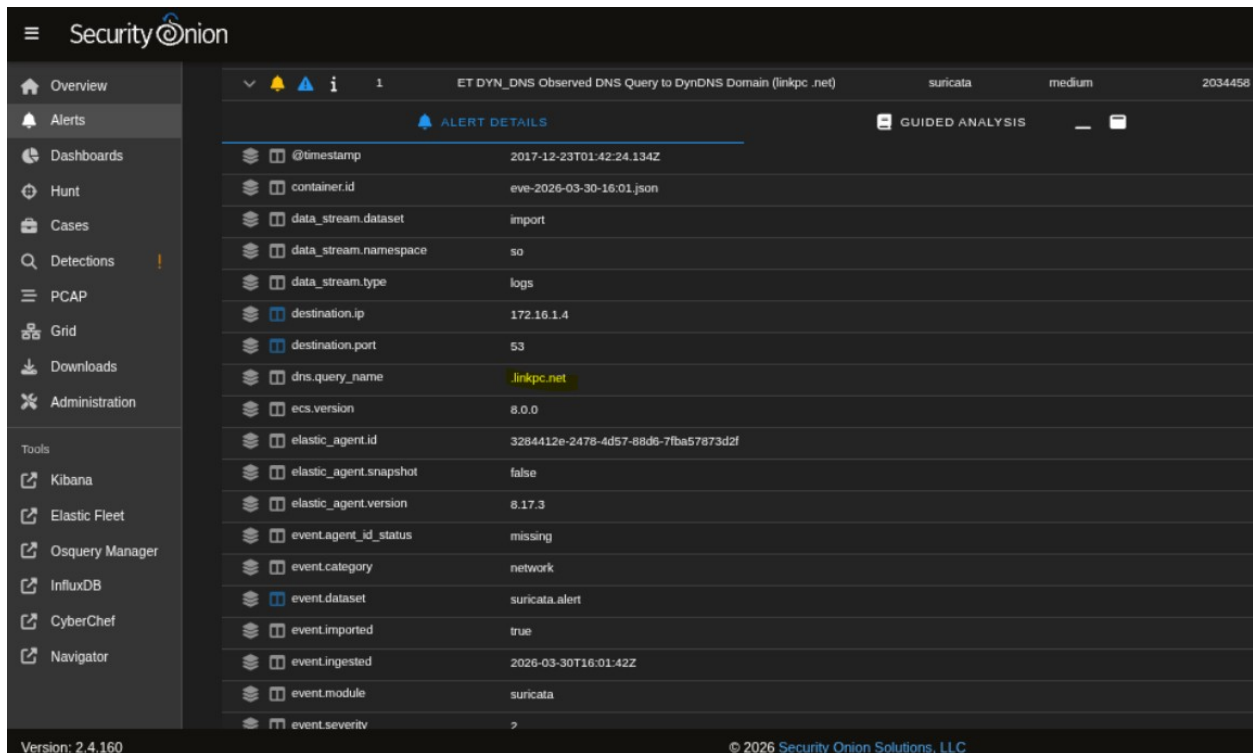


Figure 6. Figure 6: Security Onion – ET_DYN_DNS alert for .linkpc.net query. Timestamp: 2017-12-23T01:42:24Z. Source: 172.16.1.141. Destination: 172.16.1.4:53. DNS query name: .linkpc.net.

gabby4u.linkpc.net (172.16.1.141): At 01:42 UTC, host **172.16.1.141** (Jersey-Shore-1f) — a host not previously involved in malicious traffic — issued a DNS query for **gabby4u.linkpc.net**, resolving to **174.127.99.158**. This triggered ET_DYN_DNS Observed DNS Query to DynDNS Domain (linkpc.net) (SID 2034458, severity: Medium), confirming lateral movement to a new host.

Field	Value
event.severity	2
event.severity_label	medium
import.file	eve-2026-03-30-16:01.json
import.id	cc97271f8119c54e086ce75841d26dda
input.type	log
log.file.path	/hsm/import/cc97271f8119c54e086ce75841d26dda/suricata/eve-2026-03-30-16:01.json
log.id.uid	15428015316668
log.offset	143266
message	[{"timestamp":"2017-12-23T01:42:24.134664+0000","flow_id":"15428015316668","pcap_cnt":7967,"event_type":"alert","src_ip":"172.16.1.141","src_port":59986,"dest_ip":"172.16.1.4","dest_port":53,"proto":"UDP","pkt_src":"wire/pcap","community_id":"1105rAg8UUGvwhTnTNFWoNcj5JNcE","tx_id":0,"alert":{"action":"allowed","gid":1,"signature_id":2034458,"rev":4,"signature":"ET_DYN_DNS Observed DNS Query to DynDNS Domain (linkpc.net)","category":"Potentially Bad Traffic","severity":2,"metadata":{"affected_product":"Any","attack_target":"Client_Endpoint","confidence":"High","created_at":"2021_11_15","deployment":"Perimeter","former_category":"INFO","performance_impact":"Low","signature_severity":"Informational","updated_at":"2024_06_11"},"rule":"alert dns \$HOME_NET any -> any any (msg:'ET_DYN_DNS Observed DNS Query to DynDNS Domain (linkpc.net)'; dns.query; content:'linkpc.net'; nocase; endswith; content:'www.linkpc.net'; reference:url,dnsexit.com/domains/free-second-level-domains; classtype:bad-unknown; sid:2034458; rev:4; metadata:affected_product Any, attack_target Client_Endpoint, created_at 2021_11_15, deployment Perimeter, former_category INFO, performance_impact Low, confidence High, signature_severity Informational, updated_at 2024_06_11;"},"app_proto":"dns","direction":"to_server","payload_printable":".....gabby4u.linkpc.net.....","stream":0,"packet":"FP61Abw6ADbn3CkCABFAABAAARQAIArpb6sEAGNBABBOpsADUALMDonsBAAAAAAB2dhYmJ5NHUGbGlua3Bja25ldAAAAQAB","packet_info":{"linktype":1}}]
network.community_id	1105rAg8UUGvwhTnTNFWoNcj5JNcE
network.data.decoded	gabby4u.linkpc.net.....
network.packet_source	wire/pcap
network.private_ip	["172.16.1.141", "172.16.1.4"]
network.transport	UDP

Figure 7. Figure 7: Security Onion – ET_DYN_DNS alert detail showing gabby4u.linkpc.net in network.data.decoded. Source: 172.16.1.141. Transport: UDP. Confirms lateral movement to a previously clean host.

4.8 Stage 5 – RIG EK Alert Confirmation

Security Onion captured 15 instances of **ET EXPLOIT_KIT RIG EK URI Struct Mar 13 2017** (SID 2024048, severity: High), confirming active exploitation. Rule metadata confirmed malware_family: Exploit_Kit_RIG, signature_severity: Major, performance_impact: Significant, attack_target: Client_Endpoint.

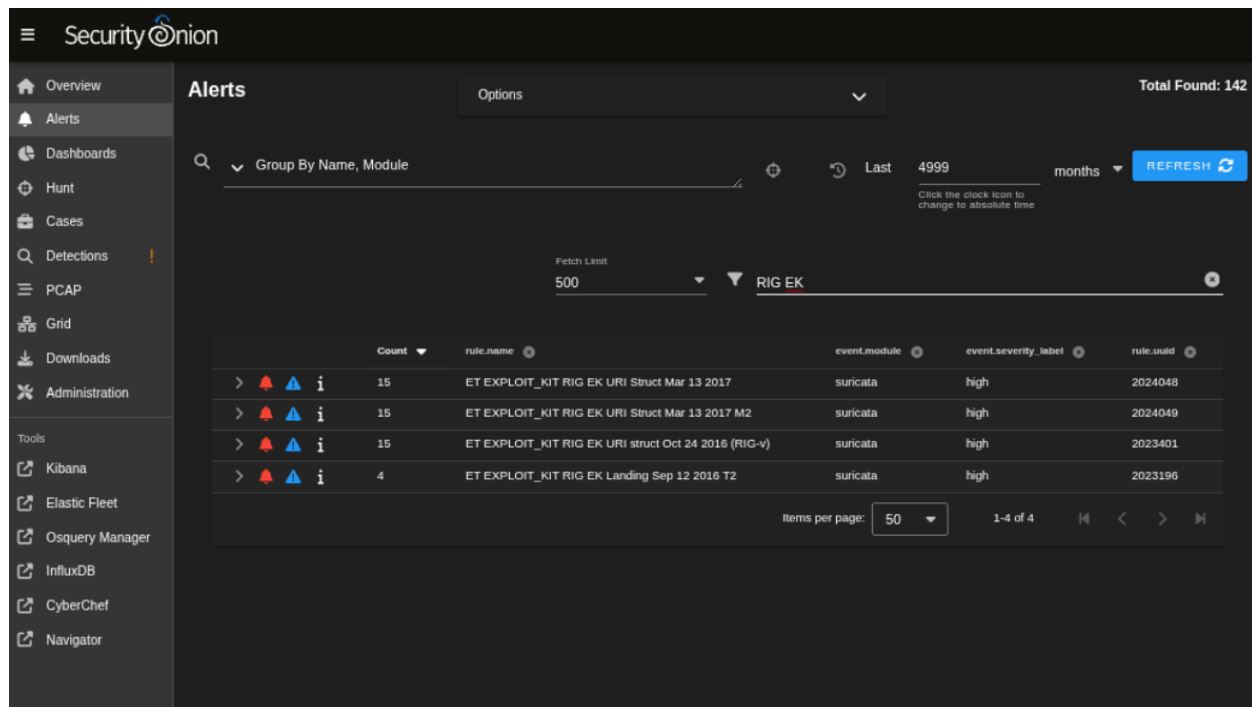


Figure 8. Figure 8: Security Onion – ET EXPLOIT_KIT RIG EK URI Struct Mar 13 2017. Count: 15 instances. Severity: High. Confirms active RIG Exploit Kit exploitation.

4.9 Stage 6 – Ransomware Deployment

Host **172.16.1.201 (Newark-1a)** exhibited definitive ransomware indicators at 01:17 UTC. DNS queries were observed for **topyzscsu5poprxy.onion.link** (a Tor2Web proxy for a ransomware payment page) and **coinurl.com** (a Bitcoin payment redirect service). These domains are opened by the ransomware in the victim browser to display the ransom note and payment instructions.

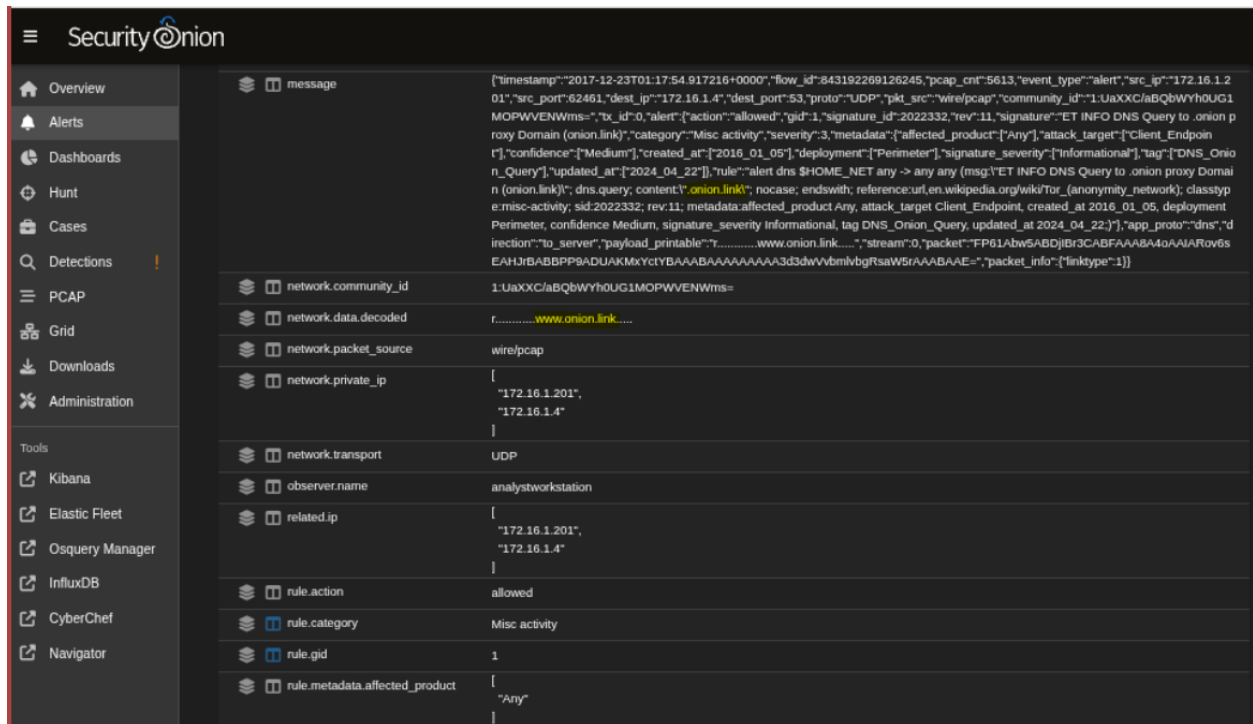


Figure 9. Figure 9: Security Onion – Alert for .onion.link DNS query. Timestamp: 2017-12-23T01:17:54Z. Source: 172.16.1.201. Confirms ransomware payment infrastructure contact.

5. Analysis and Interpretation

5.1 Files Downloaded via HTTP

NetworkMiner and Wireshark identified the following malicious files delivered to 172.16.1.67:

Table 2. Malicious Files Identified

Filename	Source	Size	Significance
Invoice[1].doc	fundtel.es (185.2.4.53)	204 KB	Malicious Word macro dropper
2042[1].exe	enjoyveganrecipes.com (78.46.96.236)	101 KB	Ransomware binary via RIG EK

5.2 C2 Infrastructure

Four distinct C2 channels were identified across the capture:

- **62.210.253.72:8080** — Primary C2. Host 172.16.1.67 sent 4 repeated POST requests with a spoofed IE7 user-agent (Dridex signature).
- **220.227.247.35:7080 / 220.227.247.39:7080** — Secondary C2 servers contacted later in the capture, confirming redundant C2 architecture.

- **153.149.153.41:8080** — Tertiary C2 contacted at end of capture.
- **gabby4u.linkpc.net (174.127.99.158)** — Dynamic DNS C2 hop used for lateral movement to 172.16.1.141.

5.3 False Positive Assessment

The probability of these alerts representing false positives is extremely low:

- **Corroborating signatures:** Multiple independent Suricata rules fired for the same traffic — RIG_EK_URI_Struct (x15), ET_DYN_DNS (linkpc.net and dyndns.org), and onion.link indicators. Independent corroboration across different rule sets eliminates coincidental triggering.
- **High confidence metadata:** All critical rules carried confidence: High and signature_severity: Major, indicating well-established, low false-positive signatures.
- **VirusTotal confirmation:** fundtel.es was flagged by Fortinet (Malware) and Forcepoint ThreatSeeker (Suspicious). C2 IP 62.210.253.72 was flagged by MalwareURL as Malware.
- **File evidence:** NetworkMiner physically extracted both malicious files (Invoice[1].doc and 2042[1].exe) from the PCAP, providing binary-level confirmation of payload delivery.
- **Dual user-agent:** The presence of a spoofed IE7 UA alongside a legitimate IE11 UA on the same host is a definitive Dridex infection indicator that cannot occur through normal browsing behaviour.

5.4 VirusTotal Corroboration

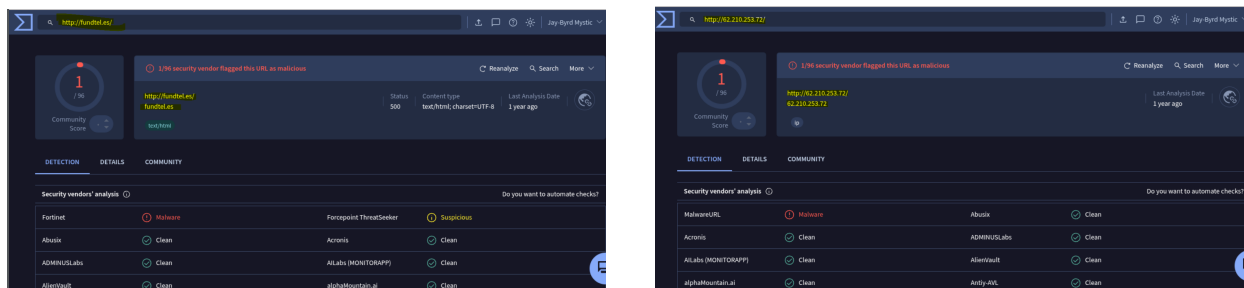


Figure 10. Figure 10: Left: VirusTotal result for fundtel.es — flagged Malware (Fortinet) and Suspicious (Forcepoint). Right: VirusTotal result for 62.210.253.72 — flagged Malware (MalwareURL). Low detection counts are expected for aged 2017 infrastructure but confirm malicious classification.

6. Conclusion

The analysis confirms that **172.16.1.67 (JERSEY-SHORE-1A)** was infected with ransomware on 2017-12-23 between 01:08 and 01:43 UTC. The infection chain began with a drive-by download of Invoice[1].doc from fundtel.es, which used a macro to download 2042[1].exe from enjoyveganrecipes.com via the RIG Exploit Kit. The installed malware established a Dridex-style C2 beacon to multiple external servers, and ransomware indicators including Tor payment pages and Bitcoin redirect services appeared on host 172.16.1.201

within 9 minutes of initial infection. Lateral movement was confirmed when 172.16.1.141 began querying `gabby4u.linkpc.net` at 01:42 UTC. A total of **four internal hosts** showed indicators of compromise within the 38-minute capture window.

7. Indicators of Compromise

IOC	Value	Type	Verdict
Patient Zero	172.16.1.67 (JERSEY-SHORE-1A)	Internal IP	Confirmed infected
Dropper Server	185.2.4.53 (fundtel.es)	External IP	Malicious (Fortinet)
Dropper File	Invoice[1].doc (204 KB)	File	Malicious Word macro
EK Server	78.46.96.236 (enjoyveganrecipes)	External IP	RIG Exploit Kit
Payload	2042[1].exe (101 KB)	File	Ransomware binary
Primary C2	62.210.253.72:8080	IP:Port	Malware (MalwareURL)
Secondary C2	220.227.247.35:7080	IP:Port	Malicious
Secondary C2	220.227.247.39:7080	IP:Port	Malicious
Tertiary C2	153.149.153.41:8080	IP:Port	Malicious
DynDNS Recon	checkip.dyndns.org	Domain	ET_DYN_DNS alert
Lateral C2	gabby4u.linkpc.net	Domain	ET_DYN_DNS alert
Ransom Page	topyzscsu5poprxy.onion.link	Domain	Tor2Web ransomware
Bitcoin Redirect	coinurl.com	Domain	Ransom payment
C2 Callback	bodyclassics.org	Domain	Payload delivery
C2 Tracker	psoeiras.net	Domain	C2 tracking
Dridex UA	Spoofed MSIE 7.0 string	User-Agent	Dridex signature
JA3 Hash	4d7a28d6...011e3 = Tofsee	TLS Hash	Malware fingerprint
MITRE T1566.002	Spearphishing Link	ATT&CK	Confirmed
MITRE T1204.002	User Execution: Malicious File	ATT&CK	Confirmed
MITRE T1071.001	Web Protocol C2	ATT&CK	Confirmed

8. Recommendations

- **Isolate and reimagine** all four affected hosts (172.16.1.67, .89, .141, .201) immediately to prevent further lateral movement or data loss.

- **Block Dynamic DNS categories** at the corporate DNS resolver. No legitimate internal application should query `checkip.dyndns.org` or `gabby4u.linkpc.net`.
- **Deploy web content filtering** to block access to known malware distribution domains at the gateway. Threat intelligence feeds would have blocked `fundtel.es` and `enjoyveganrecipes.com` before delivery.
- **Disable Office macro execution** via Active Directory Group Policy. Blocking unsigned macros in Microsoft Word would have stopped the dropper stage entirely.
- **Implement network segmentation** using VLANs or host-based firewall rules to prevent workstation-to-workstation lateral movement.
- **Alert on ransomware DNS indicators** – configure Suricata or DNS monitoring to fire immediately on `.onion.link` and `coinurl.com` DNS queries. These are high-confidence ransomware IOCs.
- **Deploy endpoint detection** with rules to detect the spoofed MSIE 7.0 user-agent from hosts whose actual browser reports a different version.
- **Block executable downloads** (`application/x-msdownload`) over unencrypted HTTP at the perimeter proxy.

9. References

- [1] Emerging Threats. *ET_DYN_DNS Ruleset*. <https://community.emergingthreats.net>
- [2] Suricata Rule SID 2034458 – *ET DYN_DNS Observed DNS Query to DynDNS Domain (linkpc.net)*. Emerging Threats, rev.4.
- [3] Suricata Rule SID 2024048 – *ET EXPLOIT_KIT RIG EK URI Struct Mar 13 2017*. Emerging Threats, rev.8.
- [4] Suricata Rule SID 2014932 – *ET INFO DynDNS CheckIp External IP Address Server Response*. Emerging Threats, rev.4.
- [5] VirusTotal. *IOC Report: fundtel.es*. <https://www.virustotal.com>
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- [10] NetworkMiner 2.8.1. *NETRESEC*. <https://www.netresec.com/?page=NetworkMiner>
- [11] Security Onion 2.4.160. *Security Onion Solutions*. <https://securityonionsolutions.com>

10. Non-Technical Summary

A staff member's computer at CARLFORCE.COM was infected with ransomware after a malicious Word document disguised as an invoice was downloaded from a compromised website. When the document was opened and macros were enabled, it silently downloaded and installed ransomware onto the machine. Once installed, the ransomware began communicating

with attacker-controlled servers and displaying payment pages demanding Bitcoin in exchange for a decryption key.

Security monitoring tools detected the attack in real time through multiple automated alerts. The infection spread indicators to three additional computers on the same internal network within 38 minutes of the first compromise.

All four affected computers should be wiped and rebuilt from clean backups immediately. Staff should be trained not to enable macros in Office documents received from unknown sources. Web filtering should be deployed to block access to known malicious websites before they can deliver harmful files. Dynamic DNS services should be blocked at the network level, as no legitimate business application within CARLFORCE.COM requires them.